Chebyshev-Exact Acceleration under Gradient Variation

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Abstract

The Chebyshev polynomial optimally contracts the initial error of a quadratic problem, but does not determine how errors in successive gradient evaluations reach the final iterate. We characterize this freedom by a factorization of the polynomial's Bezoutian and construct two realizations with constant vector memory. Among fixed-coefficient scalar linear histories with the prescribed Chebyshev terminal, the first attains the exact finite-horizon optimum for additive errors with a shared energy budget and for mean-square error under additive martingale noise. The second attains the exact finite-horizon optimum for the local response to linearly relative errors and for first-order Hessian variation at every horizon, including an explicit parity correction at odd horizons. The Hessian-variation result holds for both operator and Hilbert–Schmidt energy budgets. The lower bounds allow arbitrary real coefficients and arbitrary memory. Both methods use exactly N gradient evaluations and retain the exact Chebyshev terminal polynomial. Experiments on quantized distributed regression and consensus with a varying communication channel confirm the finite response predictions.

1. Introduction

An optimal polynomial need not have an optimal realization. On a quadratic objective, a fixed-coefficient scalar linear history with N gradient calls produces a polynomial in the Hessian applied to the initial error. If only a spectral interval is known, the smallest worst-case contraction is given by the normalized Chebyshev polynomial. Many histories produce this same terminal polynomial, while propagating errors in their gradient evaluations differently. We study this freedom under *gradient variation*: oracle evaluations may deviate from a fixed quadratic gradient through additive errors or through changes of the Hessian between calls.

The dependence on realization is classical. Ordering the reciprocal Chebyshev roots gives gradient steps with the same terminal in every order, but very different intermediate amplification. Lebedev and Finogenov constructed stable orderings [19]; these schedules returned to optimization in [1]. The Chebyshev semi-iteration of Golub and Varga [11, 12] supplies a short recurrence, whose rounding behaviour is studied in [14]. We ask: *among all fixed-coefficient linear histories with the Chebyshev terminal, what is the least possible sensitivity to gradient errors, and can it be attained with constant vector memory?*

Main results. A two-variable Bezoutian identity links each query to the transfer of its oracle error into the terminal iterate. Endpoint evaluations give sharp lower bounds, including a four-corner bound at odd horizons. Two factorizations attain them.

(1) For additive errors, Flat Chebyshev uses four physical vector slots and attains the exact terminal-distance optimum under a shared energy budget. For additive martingale noise of conditional variance at most σ^2 per call, its exact worst-case MSE is $\varepsilon_N^2 r^2 + \sigma^2(1-\varepsilon_N)^2/(\mu^2 N)$ (Theorem 3.1).

(2) For errors proportional to query distance, Balanced Chebyshev uses six slots and attains the universal local energy lower bound at every horizon, with Hessian-variation gain $A_N = 2\varepsilon_N N^{3/2}\sigma_N/(L-\mu)$, where $\sigma_N = 1$ for even N and $\sigma_N^2 = 1 + (N-1)/((N+1)N^2)$ for odd N (Theorem 4.1).

Here ε_N is the Chebyshev contraction and r bounds the initial distance. The comparison class allows arbitrary signs and arbitrary vector memory. All coefficients are computed from N and $[\mu, L]$; each method makes exactly N gradient calls. Complete proofs are in the appendices.

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Polynomial realizations and internal stability. Inexact Krylov analysis already tracks the accumulation of matrix–vector errors [28]; its Chebyshev analysis uses second-kind polynomials to control terminal errors. Internal stability of Runge–Kutta methods likewise separates the terminal stability polynomial from the propagation of stage residuals [18]. Changing a Shu–Osher representation can alter those residual responses, while the transfer of errors in the physical right-hand-side evaluations is invariant under that rewriting [18, Remark 2.4]. Our freedom is to change the underlying query history. The double-adjoint Chebyshev recurrences of Almuslimani and Vilmart [2] are a particularly close predecessor: they preserve the stability polynomial and reverse the stage construction. We use related polynomial ingredients to balance the physical responses, or the response–query products, exactly. Stochastic stabilized methods such as SRKCD [27] reuse one minibatch across the internal stages of each Runge–Kutta step, giving a different temporal noise model.

Averaging and stochastic acceleration. The asymptotically efficient covariance of averaged stochastic approximation is classical [25]. Flammarion and Bach [9] relate averaging and acceleration through a common second-order recursion; Dieuleveut, Flammarion and Bach [7] and Jain et al. [16] give accelerated least-squares guarantees, while multistage methods achieve optimal stochastic orders on general smooth strongly convex objectives [4]. Thus neither a $1/N$ variance term nor its combination with acceleration is new. Here the objective is the exact finite constant subject to a prescribed Chebyshev terminal and N fresh noisy calls. Appendix D makes this distinction explicit using the finite response formulas of the selected averaging and multistage methods.

Robustness and memory. Control-theoretic analyses quantify convergence and robustness of first-order methods through IQCs [20], stationary mean-square sensitivity [21, 29], and whole-trajectory ℓ_2 gains [13]. Function-model inexact oracles are analyzed in [5], and relative-gradient errors by frequency-domain criteria in [10]. Our gains map a shared temporal energy budget to one finite terminal response. Memory itself is also a classical resource: stationary linear iterations [3], deeper extrapolation histories [8], bundle-like gradient memory [22], low-storage Runge–Kutta schemes [17], and time-varying realization theory [6] provide different frameworks. The contribution here is an explicit constant-memory realization of the sharp finite-horizon bounds within the declared oracle model.

For Hessian variation, the sine–Jacobi method [23, Theorem 5.2] improves on the classical three-term recurrence with the same terminal; [24, Theorem 1.2] gives a nearly matching lower bound within a positive two-term class. Our Balanced construction attains the unrestricted bound at every horizon by using a longer, still constant, history. The quantitative comparison is given in Section 4. Section 5 tests both constructions on problems whose spectra and noise laws permit instance-specific predictions before simulation.

Formal verification. The theoretical results and algorithms have formal proofs in Lean. The package is available at `PolynomialRealizationsProgram/bezout`[3].

Source code. The code for the numerical experiments and figure generation is available at `PolynomialRealizationsProgram/bezout-code`[4].

2. Histories and their Bezoutian

Fix $N \geq 2$, $0 < \mu < L$, and write

$$\Delta = L - \mu, \qquad \eta = \frac{L+\mu}{\Delta}, \qquad C_s = T_s(\eta), \qquad \varepsilon_N = C_N^{-1}, \qquad n = N-1. \tag{1}$$

The Chebyshev polynomials $T_s(\cos t) = \cos(st)$ and $U_s(\cos t) = \sin((s+1)t)/\sin t$ satisfy the same three-term recurrence. In particular, $T_s(1) = 1$, $U_s(1) = s+1$, $T_N' = NU_{N-1}$, and $|T_s(z)| \leq 1$ on $[-1, 1]$.

[3] `https://github.com/dmivilensky/PolynomialRealizationsProgram/tree/main/bezout`

[4] `https://github.com/dmivilensky/PolynomialRealizationsProgram/tree/main/bezout-code`

The prescribed terminal is

$$p_N(\lambda) = \varepsilon_N T_N(\eta - 2\lambda/\Delta), \qquad \max_{\mu \le \lambda \le L} |p_N(\lambda)| = \varepsilon_N. \tag{2}$$

It is the minimax polynomial of degree at most N with $p_N(0) = 1$; see, e.g., [26].

Consider $f(x) = \frac{1}{2}(x - x_*)^T H(x - x_*)$, where $\mu I \preceq H \preceq LI$. A history has the form

$$y_s = x_0 - \sum_{j<s} a_{sj} g_j, \qquad g_s = H(y_s - x_*) + w_s, \qquad x_N = x_0 - \sum_{s<N} b_s g_s. \tag{3}$$

All scalar coefficients are fixed before the first call. They may depend on (N, μ, L) and have arbitrary signs. Let $\mathfrak{C}_N$ contain all such histories whose noiseless output is $x_* + p_N(H)(x_0 - x_*)$. Memory is unrestricted in this comparison class; only the proposed implementations have bounded storage. No additional exact Hessian products are available. Vector-slot counts include all working query and oracle-answer buffers and permit reuse of the input buffer once initialized. If an immutable copy of x_0 is retained, add one slot to each count. Scalar coefficients and the oracle's own workspace are counted separately. The analysis is in exact arithmetic, with access to the oracle $\mathcal{O}(y) = H(y - x_*) + w$. Vector norms are Euclidean; matrix norms are operator norms unless marked HS. The polynomial π_s describes the noiseless query. The *physical response* Φ_s transfers an error in the actual oracle call s to the terminal output; it does not describe a residual inserted into an internal recurrence. For a scalar polynomial a, $\|a\|_I = \max_{\lambda \in I} |a(\lambda)|$. The divided difference below is the Bezoutian of $(T_N, 1)$, up to the factor $1/2$; Bezoutians and their realization-theoretic role are classical [15]. The degree order in its factorization will encode causality of the gradient calls.

Theorem 2.1 (Realization identity)**.** *Every history in $\mathfrak{C}_N$ has nominal query polynomials π_s and physical response polynomials Φ_s such that*

$$\begin{aligned} y_s\big|_{w=0} - x_* &= \pi_s(H)(x_0 - x_*), \\ x_N - x_* &= p_N(H)(x_0 - x_*) + \sum_{s<N} \Phi_s(H) w_s, \end{aligned} \tag{4}$$

$$\sum_{s<N} \Phi_s(\lambda)\pi_s(\nu) = \frac{p_N(\lambda) - p_N(\nu)}{\lambda - \nu}. \tag{5}$$

The degrees are $\deg \pi_s = s$ *and* $\deg \Phi_s = N - 1 - s$*. The last identity extends to $\lambda = \nu$ by continuity. Set $z(\lambda) = \eta - 2\lambda/\Delta$, and define $q_s(z(\lambda)) = \pi_s(\lambda)$, $\Phi_s(\lambda) = -4\varepsilon_N f_s(z(\lambda))/\Delta$. Then*

$$B_N(x, y) := \frac{T_N(x) - T_N(y)}{2(x - y)} = \sum_{s=0}^{n} f_s(x) q_s(y), \qquad q_s(\eta) = 1. \tag{6}$$

Conversely, any graded normalized basis $\deg q_s = s$*, $q_s(\eta) = 1$, determines a unique history in $\mathfrak{C}_N$ and its unique factorization* (6)*.*

Proof. See Appendix A.

Two evaluations of the same identity give the basic lower bounds:

$$\sum_s \Phi_s(\mu) = -S_N, \qquad \sum_s \Phi_s(\mu)\pi_s(\mu) = -D_N, \qquad S_N = \frac{1 - \varepsilon_N}{\mu}, \quad D_N = \frac{2\varepsilon_N N^2}{\Delta}. \tag{7}$$

The additive problem reads the Bezoutian at $(1, \eta)$; the basic linear bound reads it at $(1, 1)$. Cauchy–Schwarz makes equal responses optimal for the first constraint, and equal response–query products optimal for the second. At odd horizons all four corners $(\pm 1, \pm 1)$ strengthen the linear bound and force a parity correction even for histories with arbitrary signs and memory. The two constructions attain the respective bounds while controlling the whole spectral interval with constant vector memory; the second balances products separately within each parity class at odd horizons.

3. Additive errors: robust and stochastic optima

Let $r, t, \sigma \geq 0$ and $\|x_0 - x_*\| \leq r$. For a history $\mathcal{R}$, define its worst-case terminal distance under a shared additive energy budget by

$$R_{\mathrm{add}}(\mathcal{R}; r, t) := \sup \left\{ \|x_N - x_*\| : \sum_{s<N} \|w_s\|^2 \leq t^2 \right\}.$$

All suprema also range over dimensions, admissible Hessians and initial points. In the stochastic model H, x_0 are deterministic, $(\mathcal{F}_s)$ is an increasing filtration, the query y_s is $\mathcal{F}_s$-measurable, the error w_s is $\mathcal{F}_{s+1}$-measurable, and

$$\mathbb{E}[w_s \mid \mathcal{F}_s] = 0, \qquad \mathbb{E}[\|w_s\|^2 \mid \mathcal{F}_s] \leq \sigma^2. \tag{8}$$

Let $R_{\mathrm{mse}}(\mathcal{R}; r, \sigma)$ be the corresponding worst-case $\mathbb{E}\|x_N - x_*\|^2$. Independence is not required, but conditional centering excludes general temporally correlated errors.

Theorem 3.1 (Exact additive optima)**.** *For every N, μ, L, r, t, σ as above,*

$$\inf_{\mathcal{R} \in \mathfrak{C}_N} R_{\mathrm{add}}(\mathcal{R}; r, t) = \varepsilon_N r + \frac{t S_N}{\sqrt{N}}, \tag{9}$$

$$\inf_{\mathcal{R} \in \mathfrak{C}_N} R_{\mathrm{mse}}(\mathcal{R}; r, \sigma) = \varepsilon_N^2 r^2 + \frac{\sigma^2 S_N^2}{N}. \tag{10}$$

Both infima are attained by the same four-slot Flat realization in Algorithm 1.

Here is its scalar preparation. Put $u_s = U_s(\eta)$, $u_{-1} = 0$, and

$$\begin{gathered} a_0 = \tfrac{1}{2} u_n, \quad a_m = u_{n-m} \ (m > 0), \quad A_m = \sum_{j=0}^{m} a_j, \quad M = A_n, \\ d = M/N, \qquad R_m = A_m - md, \qquad \tau_s = R_{n-s}/d - 1. \end{gathered} \tag{11}$$

All τ_s are nonnegative and $\tau_0 = 0$.

The construction follows directly from (6). Set

$$H_0 = 1, \quad H_m = (1 - a_m/R_m) H_{m-1} + (a_m/R_m) T_m, \qquad f_{n-m} = d H_m, \qquad q_s = (1 + \tau_s) \frac{U_s}{u_s} - \tau_s \frac{U_{s-1}}{u_{s-1}}, \tag{12}$$

where the second term is absent at $s = 0$. Each H_m is a convex combination of Chebyshev polynomials. Hence

$$\max_{\mu \leq \lambda \leq L} |\Phi_s(\lambda)| = -\Phi_s(\mu) = \frac{S_N}{N} \qquad (s < N). \tag{13}$$

This equal-response property proves both parts of Theorem 3.1; the tensor identity, feasibility and iteration are verified in Appendix B.

The balancing step also explains why the terminal is preserved. Write $Q_s = U_s/u_s$. For $1 \leq m \leq n$,

$$R_m H_m - (R_{m-1} - d) H_{m-1} = a_m T_m, \qquad d q_{n-m} = R_m Q_{n-m} - (R_m - d) Q_{n-m-1}.$$

The second identity holds also at $m = 0$; its last term is absent at $m = n$. Substitution in $\sum_m d H_m(x) q_{n-m}(y)$ telescopes to $\sum_m a_m T_m(x) Q_{n-m}(y) = B_N(x, y)$. Thus equalizing the responses changes the history without changing the Bezoutian or its terminal polynomial.

The buffer A first holds a query difference and is then overwritten by the oracle answer. Restoration uses the affine linearity of the quadratic gradient; it requires no extra oracle call.

The stochastic criterion is squared Euclidean distance with the exact terminal (2). The objective gap instead uses the H-weighted distance. Flat combines the prescribed terminal with the finite variance

Algorithm 1 Flat Chebyshev: equal additive responses

Require: x_0, $N \geq 2$, $[\mu, L]$ with $0 < \mu < L$

Scalar preparation

$n \leftarrow N-1$; $\Delta \leftarrow L-\mu$; $\eta \leftarrow (L+\mu)/\Delta$
$u_0 \leftarrow 1$, $u_1 \leftarrow 2\eta$
for $j = 2, \ldots, n$ **do**
    $u_j \leftarrow 2\eta u_{j-1} - u_{j-2}$
end for
$C \leftarrow \eta u_n - u_{n-1}$; $d \leftarrow (C-1)/(2N(\eta-1))$; $S \leftarrow u_n/2$
for $m = 0, \ldots, n$ **do**
    $\tau_{n-m} \leftarrow (S - md)/d - 1$
    $S \leftarrow S + u_{n-m-1}$ if $m < n$
end for
$\alpha_0 \leftarrow 2/(L+\mu)$, $\beta_0 \leftarrow 0$
for $s = 1, \ldots, N-2$ **do**
    $\alpha_s \leftarrow 4u_s/(\Delta u_{s+1})$; $\beta_s \leftarrow u_{s-1}/u_{s+1}$
end for
$a \leftarrow u_{n-1}/C$; $b \leftarrow 2u_n/(\Delta C)$

Oracle iteration

$V \leftarrow x_0$, $P \leftarrow x_0$, $G \leftarrow 0$
for $s = 0, \ldots, N-1$ **do**
    $A \leftarrow V - P$; $V \leftarrow V + \tau_s A$ ▷ temporary difference
    $A \leftarrow \mathcal{O}(V)$ ▷ one fresh gradient
    $V \leftarrow (V + \tau_s P)/(1+\tau_s)$
    $G \leftarrow (A + \tau_s G)/(1+\tau_s)$ ▷ affine-gradient recovery
    if $s < N-1$ **then**
        $P \leftarrow (1+\beta_s)V - \beta_s P - \alpha_s G$; swap (V, P)
    else
        $P \leftarrow (1+a)V - aP - bG$
    end if
end for
return P

constant using N noisy calls. For comparison, the classical prefix-Chebyshev recurrence has exact worst-case noise MSE $\sigma^2 V_N^{\mathrm{classical}}$, where

$$V_N^{\mathrm{classical}} = \frac{16\varepsilon_N^2}{\Delta^2}\left(\frac{N^2}{4} + \sum_{s=1}^{N-1}(N-s)^2 C_s^2\right). \tag{14}$$

At fixed μ, L this tends to a positive limit, whereas Flat gives $S_N^2/N \sim 1/(\mu^2 N)$. Averaging can also recover $1/N$ variance; the distinction here is preservation of the exact terminal.

4. Linear errors: relative robustness and Hessian variation

For errors proportional to the actual query distance, let

$$\|w_s\| \leq t\delta_s \|y_s - x_*\|, \qquad \delta_s \geq 0, \quad \sum_s \delta_s^2 \leq 1.$$

The local relative gain is

$$J_{\mathrm{rel}}(\mathcal{R}) := \limsup_{t \downarrow 0} \frac{1}{tr} \sup \|x_N - x_* - p_N(H)(x_0 - x_*)\|, \qquad r > 0. \tag{15}$$

The supremum has the same conventions as before.

Alternatively, query $g_s = (H + tE_s)(y_s - x_*)$. Let $\mathcal{T}_H(t; E)$ map $x_0 - x_*$ to $x_N(t) - x_*$, and write

$\mathcal{T}'_H[E] = \partial_t \mathcal{T}_H(0; E)$. Its spectral kernel and gain are

$$K_s(\lambda, \nu) = \Phi_s(\lambda)\pi_s(\nu), \qquad A_N(\mathcal{R}) = \max_{\lambda,\nu\in[\mu,L]} \left(\sum_{s<N} K_s(\lambda, \nu)^2 \right)^{1/2}. \tag{16}$$

For arbitrary linear perturbations E_s, this is the sharp dimension-uniform gain from the temporal Hilbert–Schmidt norm $(\sum_s \|E_s\|_{\mathrm{HS}}^2)^{1/2}$ to the Hilbert–Schmidt norm of the first variation, as in [23]. Restricting to symmetric Hessian perturbations preserves this upper bound; the Balanced construction attains it already for scalar perturbations. Define also the operator-norm gain

$$A_N^{\mathrm{op}}(\mathcal{R}) := \sup_{H,E} \left\{ \|\mathcal{T}'_H[E]\|_{\mathrm{op}} : \sum_{s<N} \|E_s\|_{\mathrm{op}}^2 \le 1 \right\}, \tag{17}$$

where the supremum includes all dimensions and admissible H. For a general history it need not equal A_N.

Theorem 4.1 (Linear energy optimum)**.** *Set*

$$\sigma_N^2 = 1 + \mathbf{1}_{N \text{ odd}} \frac{N-1}{(N+1)N^2}, \qquad G_N = \frac{D_N}{\sqrt{N}}\sigma_N = \frac{2\varepsilon_N N^{3/2}}{\Delta}\sigma_N.$$

Every $\mathcal{R} \in \mathfrak{C}_N$ satisfies

$$J_{\mathrm{rel}}(\mathcal{R}) \ge G_N, \qquad \min\{A_N(\mathcal{R}), A_N^{\mathrm{op}}(\mathcal{R})\} \ge G_N. \tag{18}$$

The six-slot Balanced realization in Algorithm 2 satisfies

$$J_{\mathrm{rel}} = A_N = A_N^{\mathrm{op}} = G_N. \tag{19}$$

Thus all three minima are attained exactly for every $N \ge 2$. The odd-horizon factor $\sigma_N = 1+O(N^{-2})$ is part of the exact optimum. The same optimum and construction coefficients hold for symmetric matrix perturbations under either matrix energy budget.

The query basis is obtained by averaging Chebyshev degrees of the same parity:

$$\Psi_0 = 1, \quad \Psi_1 = T_1, \quad \Psi_s = \Psi_{s-2} + \omega_s(T_s - T_{s-2}), \qquad Z_s = \Psi_s(\eta), \quad q_s = \Psi_s/Z_s. \tag{20}$$

Here $\omega_0 = \omega_1 = 1$ and

$$\omega_s = \begin{cases} (N+2-s-(s \bmod 2))/N, & N \text{ even}, \\ (N+1)(N+1-s)/(N^2+1), & N \text{ odd},\ s \text{ even}, \\ (N+2-s)/(N+1), & N \text{ odd},\ s \text{ odd}, \end{cases} \qquad s \ge 2. \tag{21}$$

Within each parity the ω_s decrease, so Ψ_s has nonnegative Chebyshev coefficients summing to one. The matching responses have nonnegative coefficients as well, and

$$f_s(1)q_s(1) = \begin{cases} N/2, & N \text{ even}, \\ (N^2+1)/(2(N+1)), & N \text{ odd},\ s \text{ even}, \\ (N+1)/2, & N \text{ odd},\ s \text{ odd}. \end{cases} \tag{22}$$

The common endpoint controls all spectral directions. The normalized products, rather than the responses alone, are therefore balanced.

Algorithm 2 computes all coefficients from (N, μ, L). Its reverse parity sums prepare the output weights b_s in linear scalar work; only four impulse coefficients h_{jk} per row are needed. The derivation is in Appendix C. During the iteration, v_r predicts the query at $k+r$ using the answers already observed and setting future answers to zero. Four predictions suffice.

Algorithm 2 Balanced Chebyshev: equal linear sensitivity

Require: x_0, $N \geq 2$, $[\mu, L]$ with $0 < \mu < L$

Scalar preparation: polynomial values

$n \leftarrow N - 1$; $\Delta \leftarrow L - \mu$; $\eta \leftarrow (L + \mu)/\Delta$

$C_0 \leftarrow 1$, $C_1 \leftarrow \eta$; $e_j \leftarrow 2 - \mathbf{1}_{j=0}$ for $0 \leq j \leq n$

for $j = 2, \ldots, N$ **do**

  $C_j \leftarrow 2\eta C_{j-1} - C_{j-2}$

end for

$\omega_0, \omega_1, Z_0 \leftarrow 1$; $Z_1 \leftarrow \eta$

for $s = 2, \ldots, n$ **do**

$$\omega_s \leftarrow \begin{cases} (N + 2 - s - (s \bmod 2))/N, & N \text{ even}, \\ (N+1)(N+1-s)/(N^2+1), & N \text{ odd},\ s \text{ even}, \\ (N+2-s)/(N+1), & N \text{ odd},\ s \text{ odd} \end{cases}$$

  $Z_s \leftarrow Z_{s-2} + \omega_s(C_s - C_{s-2})$

end for

Output weights

for $\sigma = 0, 1$ **do**

  $P \leftarrow 0$, $Q \leftarrow 0$

  for $s = \sigma + 2\lfloor (n - \sigma)/2 \rfloor, \ldots, \sigma$ in steps of -2 **do**

    $P \leftarrow P + e_{n-s} C_{n-s}$; $R \leftarrow Q + e_s P/2$

    $b_s \leftarrow \dfrac{4Z_s}{\Delta C_N}\left(\dfrac{R}{\omega_s} - \dfrac{Q}{\omega_{s+2}}\right)$ ▷ omit the second term if $s + 2 \geq N$

    $Q \leftarrow R$

  end for

end for

Four impulse coefficients per row

Interpret $h_{jk} = 0$ for $j \leq k$; store only $1 \leq j - k \leq 4$

$h_{10} \leftarrow -2/\Delta$

$h_{20}, h_{21} \leftarrow -4\eta\omega_2/\Delta$ if $N > 2$

$h_{30} \leftarrow -2/\Delta + \omega_3[-8\eta^2/\Delta + 4/(\Delta\omega_2)]$ if $N > 3$

$(h_{31}, h_{32}) \leftarrow (-8\eta^2\omega_3/\Delta, -4\omega_3 Z_2/(\Delta\omega_2))$ if $N > 3$

for $j = 4, \ldots, n$ **do**

  $a_j \leftarrow 2\eta\omega_j/\omega_{j-1}$; $c_j \leftarrow \omega_j/\omega_{j-2}$

  for $k = \max(0, j - 4), \ldots, j - 1$ **do**

$$h_{jk} \leftarrow a_j h_{j-1,k} + (1 - c_j) h_{j-2,k} - a_j h_{j-3,k} + c_j h_{j-4,k} - \frac{4\omega_j}{\Delta\omega_{j-1}}(Z_{j-1}\mathbf{1}_{k=j-1} - Z_{j-3}\mathbf{1}_{k=j-3})$$

  end for

end for

Oracle iteration

$X \leftarrow x_0$; $v_0, v_1, v_2, v_3 \leftarrow x_0$

for $k = 0, \ldots, N - 1$ **do**

  $G \leftarrow \mathcal{O}(v_0)$; $X \leftarrow X - b_k G$

  if $k + 4 < N$ **then**

$$v_0 \leftarrow a_{k+4}\frac{Z_{k+3}}{Z_{k+4}}v_3 + (1 - c_{k+4})\frac{Z_{k+2}}{Z_{k+4}}v_2 - a_{k+4}\frac{Z_{k+1}}{Z_{k+4}}v_1 + c_{k+4}\frac{Z_k}{Z_{k+4}}v_0 + \frac{h_{k+4,k}}{Z_{k+4}}G$$

▷ one assignment, using the old predictions

  end if

  for $r = 1, 2, 3$ with $k + r < N$ **do**

    $v_r \leftarrow v_r + (h_{k+r,k}/Z_{k+r})G$

  end for

  rotate $(v_0, v_1, v_2, v_3) \leftarrow (v_1, v_2, v_3, v_0)$

end for

return X

Comparison with shorter recurrences. The classical prefix-Chebyshev recurrence has the exact gain

$$A_N^{\text{classical}} = \frac{\varepsilon_N}{\Delta}\left(4N^2 + 16\sum_{j=1}^{N-1} j^2\right)^{1/2} \sim \frac{4}{\sqrt{3}}\frac{\varepsilon_N N^{3/2}}{\Delta}. \tag{23}$$

On the scale $\varepsilon_N N^{3/2}/\Delta$, the classical constant is $4/\sqrt{3} = 2.309401\ldots$, sine–Jacobi gives $2.137936\ldots$ [23, Theorem 5.2], and Balanced gives 2 asymptotically. The reductions are 13.4% and 6.45%, respectively, at six physical slots instead of three for the two-state recurrences. The positive two-term class of [24, Theorem 1.2] is a smaller comparison class.

Why the two realizations differ. For even N and $\sigma > 0$, additive MSE optimality forces $\Phi_0(\mu) = -S_N/N$; equality in the linear lower bound forces $\Phi_0(\mu) = -D_N/N$. But $S_N > D_N$ (Appendix C). Thus no history with an even horizon can attain both exact optima. The distinction is what is balanced: additive errors balance physical responses, while relative errors balance response–query products, separately within the parity classes at odd horizons. At even horizons the fixed first query makes these demands conflict already in the first physical response.

5. Numerical experiments

The experiments test the two balances separately. Distributed regression with quantized gradients tests additive energy and full stochastic MSE; consensus with a varying channel tests local energy and stochastic Hessian variation. The examples have explicit spectra, so the lines predict the actual instances rather than substitute worst-case upper bounds for algorithmic errors. The finite formulas and their derivations are collected in Appendix D.

Protocol and competitors. We set $L = 1$ and run each method independently from x_0 at

$$N \in \{64, 128, 256, 320, 384, 512, 640, 768, 960, 1024\}.$$

Each horizon uses its own coefficients and exactly N fresh oracle calls. Prefix is the classical Chebyshev semi-iteration. LF/fractal is the Lebedev–Finogenov ordering [19], in the recursive form of [1, Definition 5], evaluated at its dyadic horizons. For additive noise we also use unregularized accelerated averaging (DFB [7], step $1/L$, momentum one) and M-ASG [4] with $p = 1$ and first stage $\lfloor N/2 \rfloor$. For Hessian variation we use sine–Jacobi [23]. These are fixed configurations; the comparison does not optimize over their tuning families. Only M-ASG uses its prescribed stage resets.

Predictions are computed before simulation, on a grid spaced by 16 calls. Stochastic points average 2048 independent realizations; 95% intervals for ratios use 500 paired bootstrap resamples, with the same underlying random tape across methods. Deterministic energy points use the maximizing perturbation, and have no sampling error. All parameters and seeds are fixed in `bezout-code/`. The vector iterations use float64; Flat and Balanced scalar coefficients are prepared with 80 decimal digits. Balanced is evaluated through the normalized form of (37), with compensated terminal accumulation, which implements the same history as Algorithm 2.

5.1. Distributed regression with quantized gradients

Take $d = 64$, $m = 4$ workers, $\mu = 1/\kappa$, and

$$v = d^{-1/2}\mathbf{1}, \qquad H = \mu I + (1-\mu)vv^T, \qquad \rho = \mu/4.$$

Worker i has the ridge objective

$$f_i(x) = \frac{\|Ax - z_i\|^2}{2d} + \frac{\rho}{2}\|x\|^2, \quad A = \sqrt{d}\,(H-\rho I)^{1/2}, \quad z_i = \sqrt{d}\,(H-\rho I)^{-1/2}c_i, \quad \frac{1}{m}\sum_i c_i = Hx_*. \tag{24}$$

Thus $\nabla f_i(x) = Hx - c_i$ and the average objective has minimizer x_*. We start at zero and fix $\|x_*\|^2 = 1$, with 90% of its energy in $v^\perp$ and 10% in v. Each worker sends the componentwise subtractive-dither

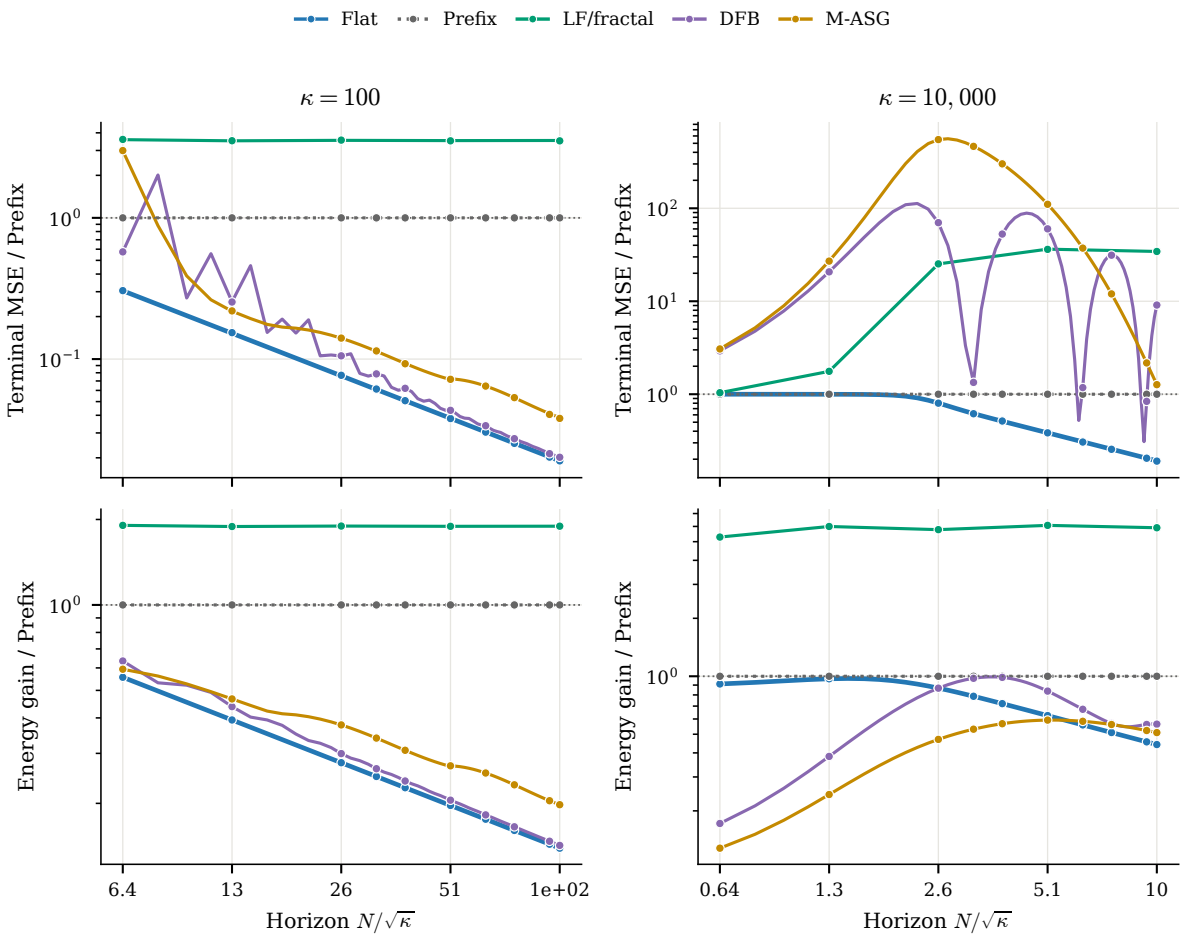


Figure 1: Distributed ridge regression with four quantized worker gradients. Top: full terminal MSE, including each method's own bias. Bottom: gain under a shared additive-energy budget. All ordinates are ratios to Prefix; both axes are logarithmic. Lines are exact finite predictions; top markers have 95% paired-bootstrap intervals, and bottom markers measure the maximizing perturbation. The horizontal coordinate is $N/\sqrt{\kappa}$. LF/fractal segments join dyadic horizons.

quantization

$$Q_h(g) = h(\mathrm{round}(g/h + u) - u), \qquad u_j \sim \mathrm{Unif}[-\tfrac{1}{2}, \tfrac{1}{2}], \qquad h = \sigma\sqrt{12m/d}.$$

Fresh independent dithers give the averaged gradient error covariance $V = \sigma^2 I/d$, independent of the query. We use $\sigma/\mu = 0.3$ and $\kappa \in \{100, 10000\}$.

For any plotted method let p be its own terminal polynomial and $V_{\mathcal{R}}(\lambda) = \sum_s \Phi_s(\lambda)^2$. Its exact MSE on this instance is

$$\mathcal{E}_{\mathcal{R}} = 0.9p(\mu)^2 + 0.1p(1)^2 + \frac{\sigma^2}{d}\big((d-1)V_{\mathcal{R}}(\mu) + V_{\mathcal{R}}(1)\big). \tag{25}$$

The deterministic additive-energy gain is $G_{\mathcal{R}} = \max_{\lambda\in\{\mu,1\}} \sqrt{V_{\mathcal{R}}(\lambda)}$. It is measured separately by starting at x_* and inserting the maximizing additive error sequence. Both quantities are divided by Prefix at the same N.

Figure 1 resolves both the stochastic constant and the accelerated transient. At $N = 1024$, Flat's MSE is 0.0190 and 0.1907 times Prefix's at $\kappa = 100$ and 10000, respectively. Against DFB the corresponding ratios are 0.943 (95% interval $[0.933, 0.954]$) and 0.0209; against M-ASG they are 0.500 and 0.150. At $\kappa = 100$, the plotted risks of Flat and DFB become close, and the finite improvement is

modest. At $\kappa = 10000$, the averaged bias produces pronounced minima in DFB's error curve, while Flat combines noise reduction with the prescribed accelerated terminal. The lower panels isolate noise sensitivity; a smaller gain alone need not give a smaller full MSE when the terminal polynomials differ.

5.2. Consensus with a varying inter-network channel

Let L_c be the Laplacian of a cycle with 16 vertices. Couple two copies through corresponding vertices and minimize $f(x) = x^T Hx/2$ over the invariant subspace $\mathcal{S}$ defined below, where

$$H = \frac{1}{4+2a}\begin{pmatrix} L_c + aI & -aI \\ -aI & L_c + aI \end{pmatrix}, \qquad a = \frac{2}{\kappa - 1}, \qquad P = \frac{1}{2}\begin{pmatrix} I & -I \\ -I & I \end{pmatrix}. \tag{26}$$

The initial disagreement lies in the invariant antisymmetric subspace $\mathcal{S} = \{(z, -z) : z \in \mathbb{R}^{16}\}$, where $P = I$, the minimizer is zero, and the extreme eigenvalues of H are $\mu = 1/\kappa$ and $L = 1$. It has squared amplitude 0.475 in each endpoint mode and total squared amplitude 0.05 spread over every other mode of $\mathcal{S}$. At call s, the oracle is

$$g_s = (H + \delta\xi_s P)y_s, \qquad \Pr(\xi_s = 1) = \Pr(\xi_s = -1) = \tfrac{1}{2}, \qquad \delta = 0.01\mu. \tag{27}$$

The signs are independent. This changes one shared inter-network conductance; all edge weights remain positive. The supplied bounds describe the reference H, as in the variation model of Section 4. We use $\kappa \in \{10000, 40000\}$.

Write λ_j for the active eigenvalues and a_j for the fixed initial amplitudes. The instance-specific insertion matrix is

$$T_{js} = a_j \Phi_s(\lambda_j)\pi_s(\lambda_j), \qquad G^{\mathrm{loc}}_{\mathcal{R}} = \|T\|, \qquad \mathbb{E}\|x_N - p_N(H)x_0\|^2 = \delta^2\|T\|^2_{\mathrm{HS}} + O(\delta^4). \tag{28}$$

The energy point differentiates the actual iteration in a unit right singular direction of T. Stochastic points instead execute the full perturbed iteration (27). Its expected terminal is exactly $p_N(H)x_0$, so these points measure noise variance. The common nominal squared error must be added to obtain full MSE.

Figure 2 shows the smaller separation predicted by the linear theory. At $N = 1024$, Balanced's variance is 0.733 and 0.759 times Prefix's; against sine–Jacobi the ratios are 0.873 ($[0.849, 0.900]$) and 0.883 ($[0.855, 0.910]$). Thus the measured variance reduction is about 12% against sine–Jacobi and 25% against Prefix. Balanced's deterministic gain ratios to Prefix and sine–Jacobi are about 0.866 and 0.935, respectively. LF/fractal has a growing loss: at $N = 1024$ its first-variation variance ratio to Prefix is 512.4 and its energy-gain ratio is about 16. In this commuting model, reordering the root steps cannot repair that loss (Proposition D.1). The conclusions concern the stated channel geometry and fixed initial signal; independently changing arbitrary edges pose a different comparison. Together the two figures verify the predicted balances, including the oscillations and crossings of the averaging baselines.

6. Conclusion

The Bezoutian identity turns the choice of a Chebyshev-exact history into a factorization problem. Flat attains the additive optima with four physical vector slots; Balanced attains the linear optimum with six, including the odd-horizon correction. Both preserve the prescribed accelerated terminal. The even-horizon incompatibility of the optima shows how the error model determines the appropriate realization. Exact finite comparisons and the two application models connect these universal bounds to physical noise sensitivity.

Use of AI tools. The authors used OpenAI Codex to assist with the Lean formalization. The authors take responsibility for the resulting proofs and code.

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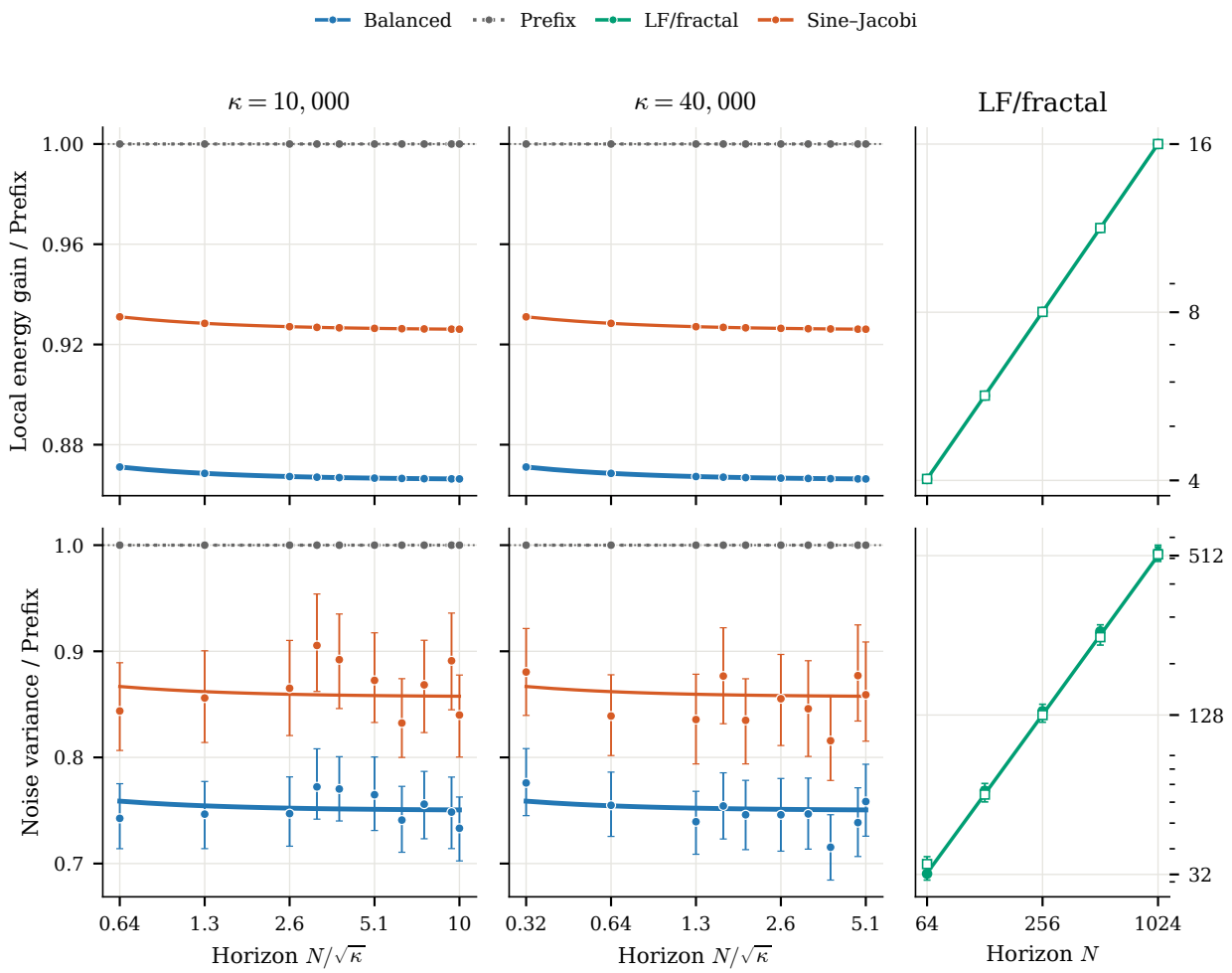


Figure 2: Consensus on two coupled cycles. Top: local temporal-energy gain. Bottom: finite-noise variance around the common Chebyshev terminal, for $\delta/\mu = 0.01$. All values are divided by Prefix. Lines use the instance-specific first variation; markers measure the differentiated iteration (top) or the full noisy iteration with 95% intervals (bottom). Left columns: linear vertical scales and logarithmic $N/\sqrt{\kappa}$. Right column: LF/fractal on logarithmic scales against N; filled circles and open squares correspond to $\kappa = 10000$ and 40000. Its normalized theoretical curve is common to both. LF/fractal segments join dyadic horizons.

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A. Realization calculus

Proof of Theorem 2.1. Let $A = (a_{sj})$ be strictly lower triangular, $b = (b_s)$, and $\mathbf{1} = (1, \dots, 1)^T$. On a scalar eigenspace,

$$\pi(\lambda) = (I + \lambda A)^{-1}\mathbf{1}, \quad \Phi(\lambda)^T = -b^T(I + \lambda A)^{-1}, \quad p_N(\lambda) = 1 - \lambda b^T(I + \lambda A)^{-1}\mathbf{1}.$$

Eliminating the gradients gives (4), including for errors depending on earlier answers. Since the two matrix inverses commute,

$$\frac{\lambda(I + \lambda A)^{-1} - \nu(I + \nu A)^{-1}}{\lambda - \nu} = (I + \lambda A)^{-1}(I + \nu A)^{-1}.$$

Multiplication by $-b^T$ and $\mathbf{1}$ proves (5), hence (6).

The coefficient of degree N in p_N is nonzero. In a strictly lower triangular N-stage history this forces $b_{N-1}\prod_{s=1}^{N-1} a_{s,s-1} \neq 0$. Hence the nominal queries have exact degrees $0, \dots, N-1$. The same unique maximal-path argument from call s to the output gives $\deg \Phi_s = N - 1 - s$. Conversely, for any such query basis with $\pi_s(0) = 1$, expand $(1 - \pi_s(\lambda))/\lambda$ uniquely in $\pi_0, \dots, \pi_{s-1}$ to obtain a_{sj}; expand $(1 - p_N(\lambda))/\lambda$ in the full basis to obtain b. This constructs the unique history. For each fixed x, expansion of $B_N(x, \cdot)$ in the query basis is unique, so its responses are exactly the proposed factors. Finally, setting $\nu = 0$ and then $\lambda = \nu = \mu$ proves (7). □

B. Additive balancing and its implementation

Proof of Theorem 3.1. The generating functions

$$\sum_{k\geq 0} U_k(y)t^k = \frac{1}{1 - 2yt + t^2}, \qquad \frac{1}{2} + \sum_{m\geq 1} T_m(x)t^m = \frac{1 - t^2}{2(1 - 2xt + t^2)}$$

give, by coefficient comparison, the divided-difference expansion

$$B_N(x, y) = \tfrac{1}{2}U_n(y) + \sum_{m=1}^{n} T_m(x)U_{n-m}(y) \tag{29}$$

At $y = \eta$ it gives the coefficients a_m in (11) and

$$M = B_N(1, \eta) = \frac{C_N - 1}{2(\eta - 1)}, \qquad \frac{4\varepsilon_N M}{\Delta} = S_N.$$

We first verify feasibility of equal masses $d = M/N$. Since $T_N' = NU_n$ and U_n is strictly increasing on $[1, \eta]$,

$$C_N - 1 = N\int_1^{\eta} U_n(t)\, dt < N(\eta - 1)u_n, \quad \text{so } a_0 > d.$$

The remaining $a_m = u_{n-m}$ decrease with m. Thus $A_m - (m+1)d$ is discretely concave for $m = 0, \dots, n$, positive at zero and zero at n. It is nonnegative throughout. Consequently $R_m \geq d$ and $R_m \geq a_m > 0$. This proves both $\tau_s \geq 0$ and the convexity assertion for H_m.

Write $Q_s = U_s/u_s$. The recursions give

$$R_m H_m = (R_{m-1} - d)H_{m-1} + a_m T_m, \quad dq_{n-m} = R_m Q_{n-m} - (R_m - d)Q_{n-m-1}.$$

The predecessor term in the first identity is omitted at $m = 0$; the final subtracted term in the second vanishes at $m = n$. Multiplying and telescoping yields

$$\sum_m dH_m(x)q_{n-m}(y) = \sum_m a_m T_m(x)Q_{n-m}(y) = B_N(x, y).$$

By Theorem 2.1, these are the physical responses of the constructed history. Convexity gives $|H_m| \le 1$ and $H_m(1) = 1$, which proves (13).

For deterministic errors, (4) and Cauchy–Schwarz give $\|x_N - x_*\| \le \varepsilon_N r + tS_N/\sqrt{N}$. For every competing history, take $H = \mu I$ and align the errors with an initial error of norm r. Its supremum is at least

$$\varepsilon_N r + t\left(\sum_s \Phi_s(\mu)^2\right)^{1/2} \ge \varepsilon_N r + tS_N/\sqrt{N},$$

by (7). This proves (9).

Under (8), all cross terms in the square of (4) vanish: earlier errors are measurable before later calls and the response matrices are deterministic. Thus

$$\mathbb{E}\|x_N - x_*\|^2 = \|p_N(H)(x_0 - x_*)\|^2 + \sum_s \mathbb{E}\|\Phi_s(H)w_s\|^2 \le \varepsilon_N^2 r^2 + \sigma^2 S_N^2/N.$$

For the lower bound use $H = \mu I$, an initial error of norm r, and independent scalar errors $\pm\sigma$ with equal probabilities in that direction. The MSE is $\varepsilon_N^2 r^2 + \sigma^2 \sum_s \Phi_s(\mu)^2$, proving (10). If both spectral endpoints must occur, take $H = \mathrm{diag}(\mu, L)$ and use only the first eigendirection. □

Verification of Algorithm 1. In the absence of errors the virtual states have residuals $U_s(z)/u_s$ and $U_{s-1}(z)/u_{s-1}$. The extrapolated query therefore has residual q_s in (12). Restoration recovers V, and affine linearity of the exact gradient gives $(\nabla f(Y) + \tau_s \nabla f(P))/(1 + \tau_s) = \nabla f(V)$. The stepsizes and momenta in Algorithm 1 implement the normalized U recurrence, including the startup $s = 0$. At the terminal step $T_N(z) = zU_{N-1}(z) - U_{N-2}(z)$ gives (2). The graded query basis and terminal determine the history uniquely, so the factorization gives its responses for every oracle-answer stream, including noisy ones. The in-place version uses exactly (V, P, G, A). No additional oracle calls are made. □

Classical additive comparison. By symmetry of (29), the prefix basis $q_s = T_s/C_s$ has responses $f_0 = U_n/2$ and $f_s = C_s U_{n-s}$ for $s > 0$. Each has its maximum modulus at one. The martingale calculation above therefore gives (14), attained at $H = \mu I$. Writing $\eta = \cosh\theta$, one has $\varepsilon_N C_{N-k} \to e^{-k\theta}$ and $\varepsilon_N C_{N-k} \le 2e^{-k\theta}$. Dominated convergence shows $V_N^{\mathrm{classical}} \to 16\Delta^{-2}\sum_{k\ge1} k^2 e^{-2k\theta} > 0$. Moreover, $\|f_s\|_{[-1,1]}\|q_s\|_{[-1,1]}$ equals $N/2$ at $s = 0$ and $N - s$ otherwise, with simultaneous attainment at $x = y = 1$. Squaring and summing these products proves (23); the same endpoint also attains the operator-norm gain bound.

C. Parity balancing and its implementation

Lemma C.1 (Separation of two masses)**.** *Let $N \ge 3$ be odd, $A = (N^2+1)/4$, and $B = (N^2-1)/4$. Then*

$$\min_{\substack{u_s, v_s \ge 0 \\ \sum_s u_s = A,\ \sum_s v_s = B}} \sum_{s=0}^{N-1} (u_s^2 + v_s^2 + 4u_s v_s) = \frac{A^2}{(N+1)/2} + \frac{B^2}{(N-1)/2} = \frac{N^3}{4}\sigma_N^2. \tag{30}$$

The same minimum holds for real u_s, v_s if the summand is replaced by

$$h(u,v) = \begin{cases} u^2 + v^2 + 4uv, & uv \ge 0, \\ u^2 + v^2 + uv, & uv < 0. \end{cases}$$

Proof. First consider the signed problem. Since $h(u,v) \ge (u^2+v^2)/2$, a minimizer exists. The function h is convex in either variable separately. Coordinatewise constrained minimization therefore gives multipliers λ, ρ for the two mass constraints. At a position with $u, v \ne 0$, the stationarity equations imply

$$2\lambda - \rho = \begin{cases} 6v, & uv > 0, \\ 3u, & uv < 0. \end{cases}$$

Thus $2\lambda - \rho$ has the strict sign of u at every such position. At a position with $u \neq 0, v = 0$, one has $\lambda = 2u$ and ρ lies between u and $4u$ by the two one-sided derivatives. Hence $2\lambda - \rho$ has the weak sign of u there. If both positive and negative u masses occurred, these conditions would force $2\lambda - \rho = 0$ and $v = 0$ at every position with $u \neq 0$. But then $\lambda = 2u$ precludes opposite signs of u. Since $\sum_s u_s = A > 0$, all u_s are nonnegative. Interchanging the variables proves the same for v_s. The signed problem therefore reduces to the nonnegative one.

The feasible set is compact. Choose a minimizer with the fewest *mixed* positions, meaning $u_s, v_s > 0$. No position can have $u_s = v_s = 0$: splitting a pure positive mass between its position and an empty one strictly decreases its square; moving the entire u mass of a mixed position to an empty one strictly decreases the cross term. There cannot be two mixed positions i, j. The feasible variation

$$(u_i, u_j, v_i, v_j) \longmapsto (u_i + t, u_j - t, v_i - t, v_j + t)$$

has quadratic coefficient -4, contradicting local minimality.

Suppose there is one mixed position (u, v), together with m pure u positions and l pure v positions, where $m + l = N - 1$. If $m, l \geq 1$, vary u by h, distribute $-h/m$ to the pure u positions, vary v by k, and distribute $-k/l$ to the pure v positions. The quadratic part is

$$(1 + 1/m)h^2 + 4hk + (1 + 1/l)k^2.$$

Its determinant $(1 + 1/m)(1 + 1/l) - 4$ is nonpositive, with equality only when $m = l = 1$. A negative determinant contradicts local minimality. In the equality case the first variation vanishes and the objective is constant along a null direction until a positive mass reaches zero. The resulting minimizer either has fewer mixed positions or has an empty position, both contradictions. If $m = 0$, all u mass is at the mixed position. For a fixed value v there, the other v masses minimize their squares by being equal, so the objective is at least

$$A^2 + v^2 + 4Av + \frac{(B - v)^2}{N - 1}.$$

This expression strictly increases for $v \geq 0$, since $B < 2(N - 1)A$. Thus $v = 0$ improves the objective. The case $l = 0$ is identical, using $A < 2(N - 1)B$. Consequently a minimizer has disjoint nonempty supports. If the u support has size m, Cauchy–Schwarz gives the minimum $A^2/m + B^2/(N - m)$, attained by equal masses on each support. The real minimizer in m is $NA/(A + B) = N/2 + 1/(2N)$, so the integer minimizer is one of $(N-1)/2, (N+1)/2$. The latter is better because $A > B$. Substitution proves (30). □

Lemma C.2 (Four-corner bound)**.** *For odd $N \geq 3$, every real separated factorization $B_N(x, y) = \sum_{s=0}^{N-1} f_s(x) q_s(y)$ satisfies*

$$\frac{3}{8} \sum_s [f_s(1)^2 q_s(1)^2 + f_s(-1)^2 q_s(-1)^2] + \frac{1}{8} \sum_s [f_s(1)^2 q_s(-1)^2 + f_s(-1)^2 q_s(1)^2] \geq \frac{N^3}{4} \sigma_N^2. \tag{31}$$

No degree, sign, or normalization conditions on the factors are needed. In fact, the stronger estimate

$$\frac{3}{8} \sum_s [f_s(1)^2 q_s(1)^2 + f_s(-1)^2 q_s(-1)^2] + \frac{1}{16} \sum_s [f_s(1) q_s(-1) + f_s(-1) q_s(1)]^2 \geq \frac{N^3}{4} \sigma_N^2 \tag{32}$$

also holds.

Proof. For odd N, the four values of B_N are $N^2/2$ on the diagonal corners and $1/2$ on the off-diagonal corners. Define

$$a_s = \frac{f_s(1) + f_s(-1)}{2}, \quad b_s = \frac{f_s(1) - f_s(-1)}{2}, \quad c_s = \frac{q_s(1) + q_s(-1)}{2}, \quad d_s = \frac{q_s(1) - q_s(-1)}{2},$$
$$\alpha_s = a_s c_s, \qquad \beta_s = b_s d_s.$$

The four corner identities give $\sum_s \alpha_s = A$ and $\sum_s \beta_s = B$, with A, B as in Lemma C.1. The contribution of position s to the left-hand side of (32) is exactly

$$\widetilde{W}_s = \alpha_s^2 + \beta_s^2 + \alpha_s\beta_s + \frac{3}{4}(a_s d_s + b_s c_s)^2 \geq h(\alpha_s, \beta_s).$$

When $\alpha_s\beta_s \geq 0$, use $(a_s d_s + b_s c_s)^2 \geq 4(a_s d_s)(b_s c_s) = 4\alpha_s\beta_s$; otherwise nonnegativity of the square suffices. The signed form of Lemma C.1 proves (32). Finally, $(u+v)^2 \leq 2(u^2+v^2)$ gives (31). □

Proof of Theorem 4.1. The scalar case $H = \mu I$ and (7) imply

$$\left(\sum_s [\Phi_s(\mu)\pi_s(\mu)]^2\right)^{1/2} \geq D_N/\sqrt{N}.$$

This proves the required bound for even N, already for scalar symmetric perturbations. For odd N, use Lemma C.2 and the scaling $\Phi_s = -4\varepsilon_N f_s/\Delta$. Writing $V_{\lambda\nu} = \sum_s K_s(\lambda,\nu)^2$, this gives

$$\frac{3}{8}(V_{\mu\mu} + V_{LL}) + \frac{1}{8}(V_{\mu L} + V_{L\mu}) \geq G_N^2. \tag{33}$$

In particular, $A_N \geq G_N$.

Every $V_{\lambda\nu}^{1/2}$ is also a lower bound for A_N^{op} and J_{rel}. For $\lambda \neq \nu$, take $H = \mathrm{diag}(\lambda, \nu)$, initial error re_2, and

$$E_s = e_s \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \qquad \sum_s e_s^2 = 1.$$

The first variation applied to e_2 is $e_1 \sum_s K_s(\lambda,\nu) e_s$, whose maximum is $V_{\lambda\nu}^{1/2}$. These symmetric perturbations have $\sum_s \|E_s\|_{\mathrm{op}}^2 = 1$, and the errors $w_s = tE_s(y_s - x_*)$ satisfy the relative-error constraint with $\delta_s = |e_s|$. The diagonal case uses scalar perturbations. Thus the maximum of the four corner energies bounds both gains from below, proving (18).

For the symmetric Hilbert–Schmidt budget, the diagonal gains are again at least $V_{\mu\mu}^{1/2}$ and $V_{LL}^{1/2}$. Choosing $E_s = e_s(e_1 e_2^T + e_2 e_1^T)/\sqrt{2}$ and projecting the output onto the unit Hilbert–Schmidt matrix $(e_1 e_2^T + e_2 e_1^T)/\sqrt{2}$ gives gain at least

$$\left(\sum_s \left[\frac{K_s(\mu, L) + K_s(L, \mu)}{2}\right]^2\right)^{1/2}.$$

The stronger inequality (32), with the same physical scaling, is a convex combination of these three squared lower bounds. It proves the symmetric Hilbert–Schmidt lower bound G_N as well.

To construct the factors, let $e_0 = 1$, $e_j = 2$ for $j > 0$. Expansion (29) and $U_k = 2T_k + 2T_{k-2} + \cdots$, with constant coefficient one when k is even, show that the coefficient of $T_a(x)T_b(y)$ in B_N is $e_a e_b/2$ when $a + b \leq n$ and $a + b \equiv n \pmod 2$, and zero otherwise. For $\sigma \in \{0, 1\}$ put

$$b_i = \sigma + 2i, \quad a_l = n - \sigma - 2l, \quad 0 \leq i, l < m_\sigma, \qquad C_{il}^\sigma = \tfrac{1}{2} e_{b_i} e_{a_l} \mathbf{1}_{i \leq l},$$

where m_σ is the number of indices $s < N$ of parity σ. Here b_i denotes a degree, not an output weight. Let $m_i = \sum_l C_{il}^\sigma$, $S_j = \sum_{i \geq j} m_i$, $\bar{m}_\sigma = S_0/m_\sigma$, and $d_j = S_j/((m_\sigma - j)\bar{m}_\sigma)$. Writing $m = m_\sigma$ and $\chi_\sigma = \mathbf{1}_{N+\sigma \text{ odd}}$, these sums are

$$m_i = e_{b_i}(m - i - \chi_\sigma/2), \qquad S_j = (m-j)(m-j+1-\chi_\sigma) - \mathbf{1}_{\sigma=j=0}(m - \chi_\sigma/2).$$

For the upper triangular all-ones matrix J factor

$$U = J^{-1}\,\mathrm{diag}(d)J, \qquad V = U^{-1}C^\sigma, \qquad C^\sigma = UV.$$

Explicitly, $U_{jj} = d_j$, $U_{ij} = d_i - d_{i+1}$ for $i < j$, and zero otherwise. Its columns sum to one. With $R_l(j) = \sum_{i \geq j} C_{il}^\sigma$,

$$V_{jl} = \frac{R_l(j)}{d_j} - \frac{R_l(j+1)}{d_{j+1}}, \tag{34}$$

where the second term is omitted at the last row. Its row sums are $S_j/d_j - S_{j+1}/d_{j+1} = \bar{m}_\sigma$. Summing the arithmetic sequences in the m_i gives $d_j = \omega_{\sigma+2j}$ exactly as in (21). The columns of U therefore give $\Psi_{\sigma+2j}$.

Both factors are nonnegative. This is immediate for U from (21). For $j \geq 1$ and $l \geq j$, $R_l(j) = e_{a_l}(l-j+1)$; the displayed d_j satisfy $d_{j+1}/d_j \geq (m_\sigma - j - 1)/(m_\sigma - j)$, hence (34) is nonnegative. The same argument applies to the odd block at $j = 0$. In the even block, $R_l(0) = e_{a_l}(l + 1/2)$ and $R_l(1) = e_{a_l} l$. Here $d_0 = 1$ and the required inequality is $d_1 \geq l/(l + 1/2)$. For even N, $d_1 = 1$; for odd N, with $m = m_0$, it is

$$d_1 = \frac{2m(m-1)}{2m^2 - 2m + 1} \geq \frac{2(m-1)}{2m-1} \geq \frac{l}{l+1/2}.$$

The case $m = 1$ has no such off-diagonal entry.

Define $F_{\sigma+2j}(x) = \sum_l V_{jl} T_{a_l}(x)$ and $f_s = Z_s F_s$. The factorization $C^\sigma = UV$ gives $B_N = \sum_s F_s \otimes \Psi_s = \sum_s f_s \otimes q_s$. Nonnegativity implies $\|f_s\|_{[-1,1]} = Z_s \bar{m}_\sigma$ and $\|q_s\|_{[-1,1]} = 1/Z_s$. The class masses are

$$\bar{m}_0 = \bar{m}_1 = N/2 \quad (N \text{ even}); \qquad \bar{m}_0 = \frac{N^2+1}{2(N+1)}, \quad \bar{m}_1 = \frac{N+1}{2} \quad (N \text{ odd}).$$

This proves (22) and $A_N = (4\varepsilon_N/\Delta)(\sum_\sigma m_\sigma \bar{m}_\sigma^2)^{1/2}$. Summation gives (19).

The same value controls arbitrary relative errors. Fix $t_0 > 0$ and write $y_s^0 - x_* = \pi_s(H)(x_0 - x_*)$ for the nominal query. Define finite scalar majorants recursively by

$$\mathsf{c}_s = 1 + (L + t_0) \sum_{j<s} |a_{sj}| \mathsf{c}_j, \qquad \mathsf{d}_s = \sum_{j<s} |a_{sj}| (L \mathsf{d}_j + \mathsf{c}_j).$$

For $0 \leq t \leq t_0$, the triangular query equations and $\delta_j \leq 1$ give, by induction, $\|y_s - x_*\| \leq r\mathsf{c}_s$ and $\|y_s - y_s^0\| \leq tr\mathsf{d}_s$. These estimates are uniform over dimensions, admissible H, and all errors, including errors chosen from the preceding history. Set $P_s = \|\Phi_s\|_{[\mu,L]}$ and $Q_s = \|\pi_s\|_{[\mu,L]}$. The response identity then gives

$$\|x_N - x_* - p_N(H)(x_0 - x_*)\| \leq tr \left(\sum_s P_s^2 Q_s^2 \right)^{1/2} + t^2 r \left(\sum_s P_s^2 \mathsf{d}_s^2 \right)^{1/2}.$$

In particular, the remainder constant depends only on the fixed finite history, L, and t_0, not on the dimension or errors. For this positive factorization the norms are attained at μ simultaneously. Scalar perturbations $w_s = te_s(y_s - x_*)$ at $H = \mu I$, with unit profile (e_s) proportional to $(\Phi_s(\mu)\pi_s(\mu))$, attain the leading coefficient. Thus J_{rel} equals the value in (19) at both even and odd horizons. □

Scalar preparation and verification of Algorithm 2. The factorization above gives the output weights explicitly:

$$b_{\sigma+2j} = \frac{4\varepsilon_N Z_{\sigma+2j}}{\Delta} \sum_l V_{jl} C_{a_l}, \tag{35}$$

since $b_s = -\Phi_s(0)$. To compute these weights without forming matrices, work backwards within each parity σ. For the degrees $b_j = \sigma + 2j$, $a_j = n - b_j$ used above, set

$$P_j = \sum_{l \geq j} e_{a_l} C_{a_l}, \qquad Q_j = \sum_{i \geq j} \frac{e_{b_i}}{2} P_i.$$

Then $P_j = P_{j+1} + e_{a_j} C_{a_j}$ and $Q_j = Q_{j+1} + e_{b_j} P_j/2$, with zero terminal sums. Multiplying (34) by C_{a_l} and summing gives

$$\sum_l V_{jl} C_{a_l} = \frac{Q_j}{\omega_{b_j}} - \frac{Q_{j+1}}{\omega_{b_j+2}},$$

where the second term is absent at the last index. These are exactly the reverse sums (P, Q, R) in Algorithm 2. All ω_s are positive, and $Z_s \geq 1$, since Ψ_s is a convex combination of T_j and $C_j \geq 1$.

The coefficients h_{jk} in the raw queries $Z_j y_j = Z_j x_0 + \sum_{k<j} h_{jk} g_k$ are zero for $j \le k$. The startup rows printed in Algorithm 2 follow from the startup equations and the first step of (37) below. For $j \ge 4$, with $a_j = 2\eta\omega_j/\omega_{j-1}$ and $c_j = \omega_j/\omega_{j-2}$, the recurrence is

$$h_{jk} = a_j h_{j-1,k} + (1 - c_j) h_{j-2,k} - a_j h_{j-3,k} + c_j h_{j-4,k} - \frac{4\omega_j}{\Delta\omega_{j-1}} (Z_{j-1} \mathbf{1}_{k=j-1} - Z_{j-3} \mathbf{1}_{k=j-3}). \tag{36}$$

Only the band $1 \le j - k \le 4$ is needed by the algorithm; its recurrence closes within that band. These coefficients and the readout weights require $O(N)$ scalar storage, not vector history. To verify it, let $Q_s = Z_s y_s$ and $W_s = (Q_s - Q_{s-2})/\omega_s$ for $s \ge 2$, with $W_1 = 0$. The startup equations are

$$Q_1 = \eta x_0 - 2g_0/\Delta, \qquad W_2 = 2(\eta^2 - 1)x_0 - 4\eta(g_0 + g_1)/\Delta.$$

The polynomial identity for $T_s - T_{s-2}$ gives, for $s \ge 2$,

$$W_{s+1} = 2\eta W_s - W_{s-1} - \frac{4}{\Delta\omega_s}(Z_s g_s - Z_{s-2} g_{s-2}), \qquad Q_{s+1} = Q_{s-1} + \omega_{s+1} W_{s+1}. \tag{37}$$

Indeed on an exact quadratic, writing $Z = \eta I - 2H/\Delta$, $W_s = (T_s(Z) - T_{s-2}(Z))(x_0 - x_*) + (C_s - C_{s-2})x_*$ for $s \ge 2$. The U recurrence and the startup equations prove (37), including its first step. These equations define a scalar linear history with the required nominal queries for any oracle stream. Eliminating W gives (36) with the startup coefficients in Algorithm 2. With future answers zero, four consecutive raw queries obey the homogeneous fourth-order recurrence. Adding the impulse of the new answer to all four future predictions gives exactly Algorithm 2. By induction these predictions equal the queries of the history defined by (37); its readout (35) is the unique one prescribed by the Bezoutian. The iteration uses $(v_0, v_1, v_2, v_3, X, G)$ and exactly one call per step.

Hessian-variation interpretation. If Π_λ are the spectral projectors of H, differentiation of the linear history gives

$$\mathcal{T}'_H[E] = \sum_{s,\lambda,\nu} K_s(\lambda, \nu) \Pi_\lambda E_s \Pi_\nu.$$

Orthogonality of the spectral blocks and Cauchy–Schwarz in s yield

$$\|\mathcal{T}'_H[E]\|^2_{\mathrm{HS}} \le A_N^2 \sum_s \|E_s\|^2_{\mathrm{HS}}.$$

For unrestricted matrix perturbations equality can be realized on a maximizing spectral pair. For Balanced, the maximizing pair is (μ, μ), so scalar, symmetric Hessian perturbations already attain the coefficient.

For the operator-norm budget, use the matrix form $\mathcal{T}'_H[E] = \sum_s \Phi_s(H) E_s \pi_s(H)$. The positive factorization constructed above satisfies $P_s Q_s = |\Phi_s(\mu)\pi_s(\mu)|$, so

$$\|\mathcal{T}'_H[E]\|_{\mathrm{op}} \le \sum_s P_s Q_s \|E_s\|_{\mathrm{op}} \le A_N \left(\sum_s \|E_s\|^2_{\mathrm{op}} \right)^{1/2}.$$

Taking $H = \mu I$ and scalar E_s proportional to $\Phi_s(\mu)\pi_s(\mu)$ gives equality. Thus $A_N^{\mathrm{op}} = A_N$ for this construction, both with unrestricted and with symmetric perturbations. The argument is deterministic and imposes no independence or temporal decorrelation assumption.

Incompatibility of the exact optima. For even N and $\sigma > 0$, equality in the additive MSE lower bound requires all $\Phi_s(\mu) = -S_N/N$; equality in the linear lower bound requires all $\Phi_s(\mu)\pi_s(\mu) = -D_N/N$. Since $\pi_0 = 1$, both would imply $S_N = D_N$. However,

$$\frac{S_N}{D_N} = \frac{T_N(\eta) - 1}{N^2(\eta - 1)} > 1.$$

Indeed $T'_N = N U_{N-1}$ and U_{N-1} strictly increases on $(1, \infty)$ for $N \ge 2$, so the secant slope exceeds $T'_N(1) = N^2$. This argument uses no restrictions on signs or memory.

D. Finite comparisons and experimental predictions

D.1. The plotted competitor configurations

The following formulas specify the competitors and compute their actual finite responses. Prefix, LF/fractal and sine–Jacobi use

$$x_{s+1} = (1+\beta_s)x_s - \beta_s x_{s-1} - \alpha_s \mathcal{O}(x_s), \qquad x_{-1} = x_0. \tag{38}$$

For Prefix, $\alpha_0 = 2/(L+\mu)$, $\beta_0 = 0$, and $\alpha_s = 4C_s/(\Delta C_{s+1})$, $\beta_s = C_{s-1}/C_{s+1}$ for $s \geq 1$. For LF/fractal, $\beta_s = 0$ and

$$\alpha_s = \left(\frac{L+\mu}{2} - \frac{\Delta}{2}\cos\frac{(\varpi_N(s+1) - 1/2)\pi}{N}\right)^{-1}.$$

Here the permutation starts with $\varpi_1 = (1)$ and doubles by interlacing ϖ_m with $2m + 1 - \varpi_m$ [1, Definition 5]. For sine–Jacobi, set $r_1 = \eta$ and

$$\begin{aligned} c_s &= \frac{\sin^2(\pi s/N)}{4\sin(\pi(s-1/2)/N)\sin(\pi(s+1/2)/N)}, \quad r_{s+1} = \eta - c_s/r_s, \\ \alpha_s &= \frac{2}{\Delta r_{s+1}}, \qquad\qquad \beta_s = \frac{c_s}{r_s r_{s+1}} \qquad (1 \leq s < N), \end{aligned} \tag{39}$$

with the same initial step as Prefix. These are the normalized Jacobi coefficients of [23]; they preserve the terminal (2).

DFB uses $y_s = 2x_s - x_{s-1}$, $x_{s+1} = y_s - L^{-1}\mathcal{O}(y_s)$, and returns $(N+1)^{-1}\sum_{j=0}^{N} x_j$ [7, Eqs. (3)–(4)]. Its extra regularization parameter is zero; the ridge term in (24) remains part of the objective. At an eigenvalue λ, define $d_{-1} = d_0 = 1$ and $d_{j+1} = (1-\lambda/L)(2d_j - d_{j-1})$. Then

$$p^{\mathrm{DFB}}(\lambda) = \frac{1}{N+1}\sum_{j=0}^{N} d_j, \qquad \Phi_s^{\mathrm{DFB}}(\lambda) = -\frac{1-d_{N-s}}{\lambda(N+1)}. \tag{40}$$

Indeed a persistent scalar error shifts the equilibrium by $-w/\lambda$; its accumulated response after k calls is $-(1-d_k)/\lambda$. Averaging the impulse at call s over its remaining states gives (40). In particular, its variance coefficient is $\sum_{j=1}^{N}(1-d_j)^2/(\lambda^2(N+1)^2)$.

M-ASG uses $y_s = (1+\beta_k)x_s - \beta_k x_{s-1}$ and $x_{s+1} = y_s - \alpha_k \mathcal{O}(y_s)$ within stage k. At each stage start the previous state is reset to the current state. We use the parameters of [4, Theorem 3.4, Corollary 3.8] with $p = 1$:

$$\begin{aligned} n_1 = \lfloor N/2 \rfloor, \quad \alpha_1 = 1/L, \quad B = \lceil 3\sqrt{\kappa}\log 2 \rceil, \\ n_k = 2^k B, \quad \alpha_k = (4^k L)^{-1}\ (k \geq 2), \qquad \beta_k = \frac{1-\sqrt{\mu\alpha_k}}{1+\sqrt{\mu\alpha_k}}. \end{aligned} \tag{41}$$

The last stage stops when the total reaches N, including an incomplete stage. This fixes a method even below the budget condition $N \geq 2\sqrt{\kappa}$ of the cited corollary; the predictions here use its recurrence rather than extend that corollary's guarantee.

For completeness, a two-by-two propagation computes every required physical response, including incomplete M-ASG stages. For (38), set

$$M_s(\lambda) = \begin{pmatrix} 1+\beta_s - \alpha_s\lambda & -\beta_s \\ 1 & 0 \end{pmatrix}.$$

For extrapolated queries replace the first row by $((1-\alpha_s\lambda)(1+\beta_s), -(1-\alpha_s\lambda)\beta_s)$. At a reset multiply M_s on the right by $J = \left(\begin{smallmatrix} 1 & 0 \\ 1 & 0 \end{smallmatrix}\right)$. With $e_1 = (1,0)^T$, the last-iterate terminal and responses are

$$p(\lambda) = e_1^T M_{N-1}\cdots M_0 \begin{pmatrix} 1 \\ 1 \end{pmatrix}, \qquad \Phi_s(\lambda) = -\alpha_s e_1^T M_{N-1}\cdots M_{s+1} e_1. \tag{42}$$

Empty products are identities. This follows by inserting one physical error into the first state component and propagating it through later steps. It also proves that the formulas concern oracle errors rather than residuals inserted into a rearranged recurrence.

D.2. Additive predictions and deterministic measurements

For independent centered additive errors of covariance V, (4), also valid with a different terminal p, gives

$$\mathbb{E}\|x_N - x_*\|^2 = \|p(H)(x_0 - x_*)\|^2 + \sum_s \operatorname{tr}\big(\Phi_s(H) V \Phi_s(H)^T\big). \tag{43}$$

Cross terms vanish by conditional centering. Subtractive dithering has component variance $h^2/12$ independent of the input; independence between workers divides it by m. This proves the covariance used in Section 5. The multiplicities $d-1$ and 1 and the initial energies 0.9 and 0.1 then give (25). Thus DFB and M-ASG are compared using their own exact finite biases, not their published upper bounds. The Flat and Balanced responses are those already derived in Appendices B and C.

In an orthonormal eigenbasis, the map from the stacked errors to the terminal separates into rows $(\Phi_s(\lambda))_s$. Its operator norm is $\max_{\lambda\in\operatorname{spec}(H)} \sqrt{\sum_s \Phi_s(\lambda)^2}$. It is attained by a single maximizing eigenvector and temporal profile $\Phi_s(\lambda)/\sqrt{\sum_j \Phi_j(\lambda)^2}$. The lower panels of Fig. 1 run this physical additive sequence, of total energy μ^2, from x_* and divide the measured error by μ.

D.3. Channel variation and its finite-noise interpretation

On $\mathcal{S}$, the eigenvalues of (26) are

$$\lambda_j = \frac{\ell_j + 2a}{4 + 2a}, \qquad \ell_j = 2 - 2\cos(2\pi j/16), \quad 0 \le j < 16.$$

For a deterministic channel profile $e = (e_s)$, differentiation of the actual oracle gives $\dot g_s = H\dot y_s + e_s P y_s^0$. Since $P = I$ on $\mathcal{S}$, (4) gives $\dot x_N = \sum_s \Phi_s(H) e_s \pi_s(H) x_0$. Its eigenbasis coordinates are exactly Te from (28). Hence the sharp unit-energy response is $\|T\|$, attained by its top right singular vector. This proves the deterministic prediction and the differentiated-iteration test.

For the finite random oracle, every fixed-coefficient history has a multilinear expansion

$$x_N = \sum_{I\subseteq\{0,\dots,N-1\}} \delta^{|I|} v_I \prod_{s\in I} \xi_s, \qquad v_\varnothing = p_N(H)x_0, \quad v_{\{s\}} = T_{\cdot s}.$$

The last equality is in the orthonormal eigenbasis. Each oracle sign enters only at its own call, so induction on the triangular history proves multilinearity. Products of distinct independent centered signs are orthogonal in L^2, whence

$$\mathbb{E}x_N = p_N(H)x_0, \qquad \mathbb{E}\|x_N - p_N(H)x_0\|^2 = \sum_{I\neq\varnothing} \delta^{2|I|}\|v_I\|^2 = \delta^2\|T\|_{\mathrm{HS}}^2 + O(\delta^4). \tag{44}$$

The remainder is nonnegative and the expansion holds at fixed N. The stochastic markers evaluate the full left-hand side. The nominal squared bias is common to all plotted methods, so it can be added to recover full MSE but would dilute a percentage comparison of variances.

The endpoint signal energies explain why the unrestricted gain improvement survives this particular network geometry. For Prefix, sine–Jacobi and Balanced the two endpoint time profiles $K_s(\mu,\mu)$ and $K_s(1,1)$ agree up to an overall sign, and their common squared norm is A_N^2. Consequently

$$\sqrt{0.95}\, A_N \le \|T\| \le A_N, \qquad 0.95 A_N^2 \le \|T\|_{\mathrm{HS}}^2 \le A_N^2. \tag{45}$$

The upper bounds follow from $\sum_j a_j^2 = 1$ and the spectral maximum principles of the three methods. For the first lower bound take the temporal direction of either endpoint profile; for the second sum their squared entries. Writing $c_{\sin} := \lim_{N\to\infty}(A_N^{\mathrm{sine}}/A_N^{\mathrm{Balanced}})^2 = 1.142692\ldots$ for the sine–Jacobi constant from [23, Theorems 5.1–5.2], we obtain the asymptotic variance separation $\|T_{\mathrm{sine}}\|_{\mathrm{HS}}^2/\|T_{\mathrm{Balanced}}\|_{\mathrm{HS}}^2 \ge$

$0.95c_{\sin} + o(1) = 1.0855\ldots + o(1)$. The curves use the exact matrices T, retaining the bulk modes and all finite-N effects.

Proposition D.1 (Root ordering in the commuting channel model)**.** *For every permutation of the Chebyshev root steps,*

$$K_s(\lambda,\lambda) = -\alpha_s \prod_{j\neq s}(1-\alpha_j\lambda) = -\frac{p_N(\lambda)}{\alpha_s^{-1}-\lambda}.$$

Permuting the roots leaves $\|T\|$ and $\|T\|_{\mathrm{HS}}$ unchanged. It also leaves the finite-noise terminal distribution unchanged for independent identically distributed channel signs. Its exact variance in (27) *is*

$$\mathcal{V}_{\mathrm{LF}}(\delta) = \sum_j a_j^2 \left\{ \prod_{s<N} \left[(1-\alpha_s\lambda_j)^2 + \delta^2\alpha_s^2\right] - p_N(\lambda_j)^2 \right\}. \tag{46}$$

For the network family above, along the dyadic horizons,

$$\frac{\|T_{\mathrm{LF}}\|}{\|T_{\mathrm{Prefix}}\|} \sim \frac{\sqrt{N}}{2}, \qquad \frac{\|T_{\mathrm{LF}}\|_{\mathrm{HS}}^2}{\|T_{\mathrm{Prefix}}\|_{\mathrm{HS}}^2} \sim \frac{N}{2}. \tag{47}$$

Proof. Multiplying the nominal prefix and physical suffix cancels their dependence on the ordering and gives the displayed kernel; at a root use its continuous polynomial extension. Reordering therefore only permutes the columns of T. For finite noise all matrices $I - \alpha_s(H + \delta\xi_s P)$ commute. Reordering the paired (α_s, ξ_s) leaves their product unchanged, and exchangeability of the signs proves the distributional statement. On mode j, the finite multiplier is $\prod_s(1 - \alpha_s\lambda_j - \delta\alpha_s\xi_s)$. Independence gives its second moment as the product in (46); subtracting the squared mean and summing the mode energies proves that formula.

Put $k_s = K_s(\mu,\mu)$ and $l_s = K_s(1,1)$ for an even horizon. Logarithmic differentiation of T_N at 1, using $T_N'(1) = N^2$, $T_N''(1) = (N^4 - N^2)/3$, gives

$$\|k\|^2 = \|l\|^2 = \frac{4\varepsilon_N^2}{3\Delta^2}(2N^4 + N^2), \qquad |\langle k, l\rangle| = \frac{4\varepsilon_N^2 N^2}{\Delta^2}.$$

The endpoint contribution to $\|T\|^2$ is therefore $0.475(\|k\|^2 + |\langle k, l\rangle|)$, and that to $\|T\|_{\mathrm{HS}}^2$ is $0.95\|k\|^2$. At any fixed interior eigenvalue the squared root-step profile is $p_N'(\lambda)^2 - p_N(\lambda)p_N''(\lambda) = O(\varepsilon_N^2 N^2)$; thus the finitely many bulk modes are lower order. For Prefix their contribution is $O(\varepsilon_N^2 N)$, while the endpoint contribution to both squared norms is $0.95(A_N^{\mathrm{classical}})^2 \sim 0.95 \cdot 16\varepsilon_N^2 N^3/(3\Delta^2)$ by (23). Taking the ratios proves (47). Finally $z(\lambda_j) = 1 - \ell_j/2$ is independent of κ. The common factor ε_N/Δ cancels from first-variation ratios to Prefix, explaining the single normalized LF curve in Fig. 2. □

At $N = 1024$, (46) exceeds the leading LF variance by 4.21% and 0.262% at $\kappa = 10000$ and 40000. The corresponding exact finite LF/Prefix variance ratios are 533.8 and 513.7, compared with the common first-variation prediction 512.4. Thus the finite-noise points can depend on κ even though the normalized local curves coincide.